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20CYS443- BIOMETRICS AND SECURITY

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TIFAC CORE in Cyber Security

Class: IV B.TECH CYS, 7th Semester

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Course Objectives

- Understand the technological uplifts with biometrics compared to traditional securing mechanisms and standards applied to security
- To understand the concepts of different types of biometrics and to enable design of biometric system and its privacy risks
- To familiarize with biometric interface and biometric applications

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Concept Map

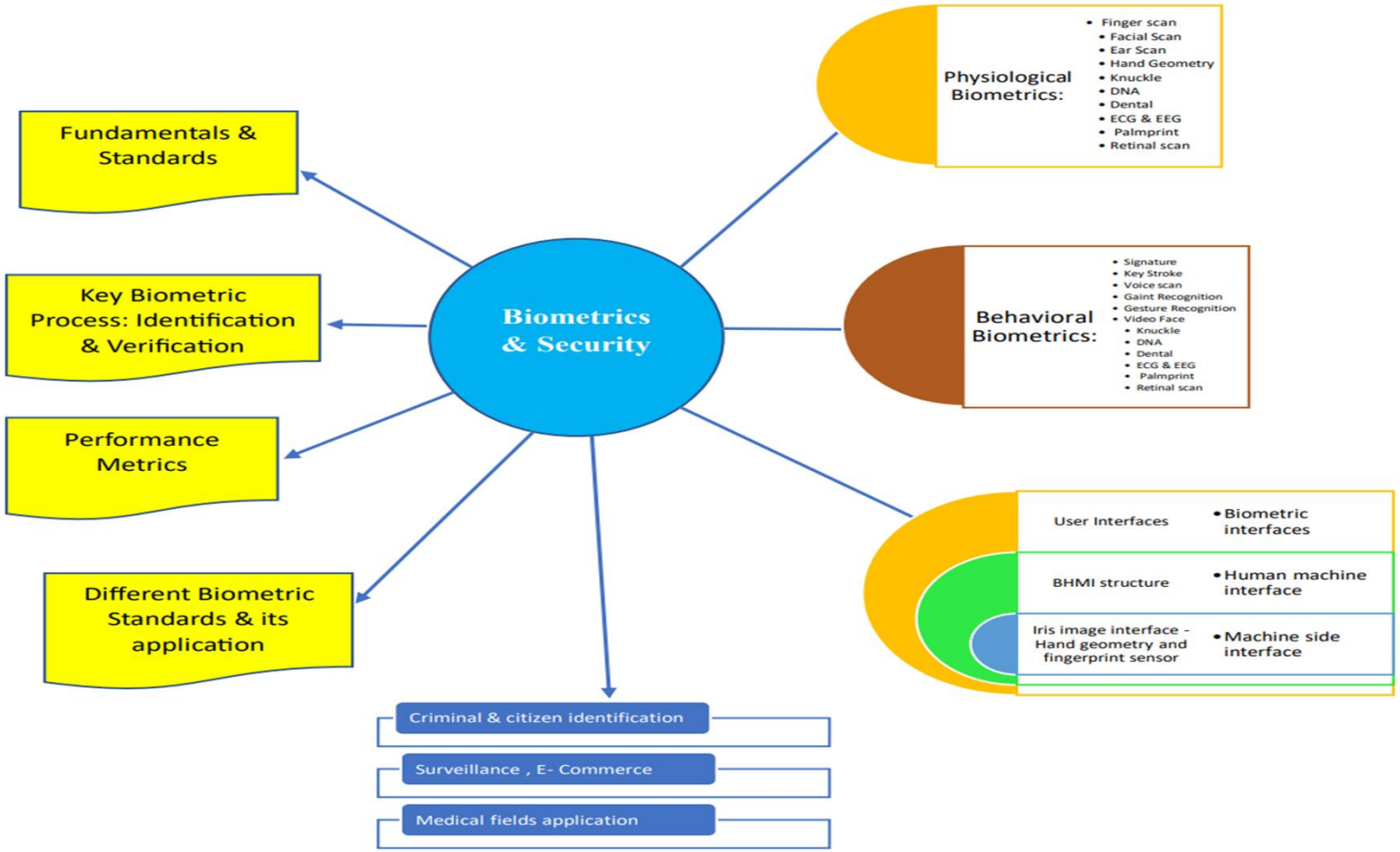


Table 2A: Internal Assessment Components

Component	TL Courses (4 & 3 Credits)	Theory Courses (3 Credits)	Lab Courses (1 Credit)
CA-1	8	7	—
CA-2 (Theory) / Lab Evaluation-1 (TL) *(including viva voce)*	7	8	20
Mid-Term Examination	30 (20 Theory + 10 Lab)	30	20
CA-3	8	7	—
CA-4 (Theory) / Lab Evaluation-2 (TL) *(including viva voce)*	7	8	20
Total Internal Assessment Marks	**60**	**60**	**60**

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Evaluation & Grading

S.No		Weightage
1.	Mid Term	30
2.	Continuous Assessment (Theory) (CAT)	30
3.	End Semester Theory	40
Total		**100**

Biometrics- Introduction

- Biometrics is the science of establishing the identity of an individual based on the physical, chemical or behavioral attributes of the person.
- Reliable authorization and authentication are becoming necessary for many everyday actions (or applications), be it boarding an aircraft, performing a financial transaction, or picking up a child from daycare
- Biometric technologies are automated methods of verifying or recognizing the identity of a living person based on a physiological or behavioral characteristic

Biometrics- Introduction

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- A biometric is a physiological or behavioral characteristic of a human being that can distinguish one person from another

- Used for identification or verification of identity

Physiological are related to the shape of the body

Example

Fingerprint
- face recognition
- DNA
- Palm print,
- hand geometry
- iris recognition,

Behavioral are related to the behavior of a person

Example

- Voice
- Signature
- Typing pattern

Why we need biometrics ?

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- Traditional means of automatic identification (before biometrics)
  - Knowledge-based
    - Use “something that you know”
    - Examples: password, PIN
  - Token-based
    - Use “something that you have”
    - Examples: credit card, smart card, keys

Why we need biometrics ?

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- Problems with traditional approaches
  - by the imposters
  - (25% of people seem to wriToken may be lost, stolen or forgotten
  - PIN may be forgotten or guessed te their PIN on their ATM card)
- Estimates of annual identity fraud damages per year:
  - $1 billion in welfare disbursements
  - $1 billion in credit card transactions
  - $1 billion in fraudulent cellular phone use
  - $3 billion in ATM withdrawals

Why we need biometrics ?

The traditional approaches are unable to differentiate between an authorized person and an imposter

⇨Use biometrics which relies on “who you are” or “what you do”

Why we need biometrics ?

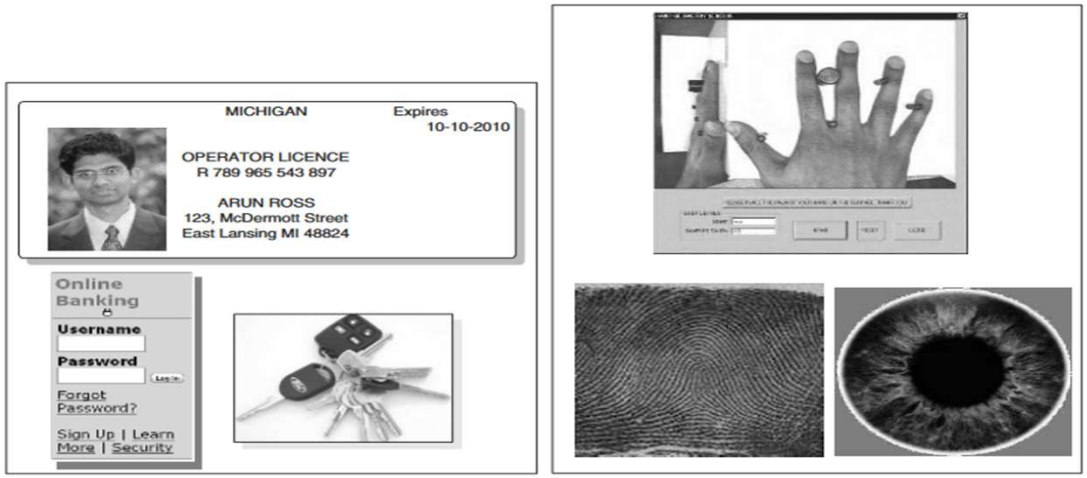
- Recognizing people is a fundamental activity at the heart of our society and culture, since for many activities (applications), ensuring the identity and authenticity of people is a prerequisite.
- Biometric identification, or biometrics, refers to identifying an individual based on his or her distinguishing characteristics.



Figure 1.1: Distinct personal characteristics: physiological (fingerprint) and behavioral characteristics (signature).

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Traditional & Biometrics Schemes



Authentication schemes. (a) Traditional schemes use ID cards, passwords and keys to validate individuals and ensure that system resources are accessed by a legitimately enrolled individual.

(b) With the advent of biometrics, it is now possible to establish an identity based on “who you are” rather than by “what you possess” or “what you remember”.

Benefits of Biometrics versus Traditional Authentication Methods

- The most frequently used authentication technologies are passwords and PINs. They secure access to personal computers (PCs), networks, and applications; control entry to secure areas of a building; and authorize automatic teller machine (ATM) and debit transactions.

- Handheld tokens (such as cards and key fobs) have replaced passwords in some higher-security applications.

Benefits of Biometrics versus Traditional Authentication Methods

Increased Security:

- Biometrics can provide a greater degree of security than traditional authentication methods, meaning that resources are accessible only to authorized users and are kept protected from unauthorized users.
- Passwords and PINs are easily guessed or compromised; tokens can be stolen.
- Biometrics data cannot be guessed or stolen in the same fashion a password or token

Benefits of Biometrics versus Traditional Authentication Methods

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Increased Convenience:

- One of the reasons passwords are kept simple (and are then subject to compromise) is that they are easily forgotten.
- biometrics are difficult if not impossible to forget, they can offer much greater convenience than systems based on remembering multiple passwords or on keeping possession of an authentication token.
- Highly sensitive information can more readily be made available on a biometrically protected network than on one protected by passwords.

Benefits of Biometrics versus Traditional Authentication Methods

Increased Accountability:

- Given the increased awareness of security issues in the enterprise and in customer-facing applications, the need for strong auditing and reporting capabilities has grown more pronounced.
- Using biometrics to secure computers and facilities eliminates phenomena such as buddy-punching and provides a high degree of certainty as to what user accessed what computer at what time.

- Even if the auditing and reporting capabilities of a system are rarely used, the fact that they exist often serves as an effective deterrent.

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Modes of Authentications

1. *Possessions:* Physical possessions such as keys, passports, and smartcards.

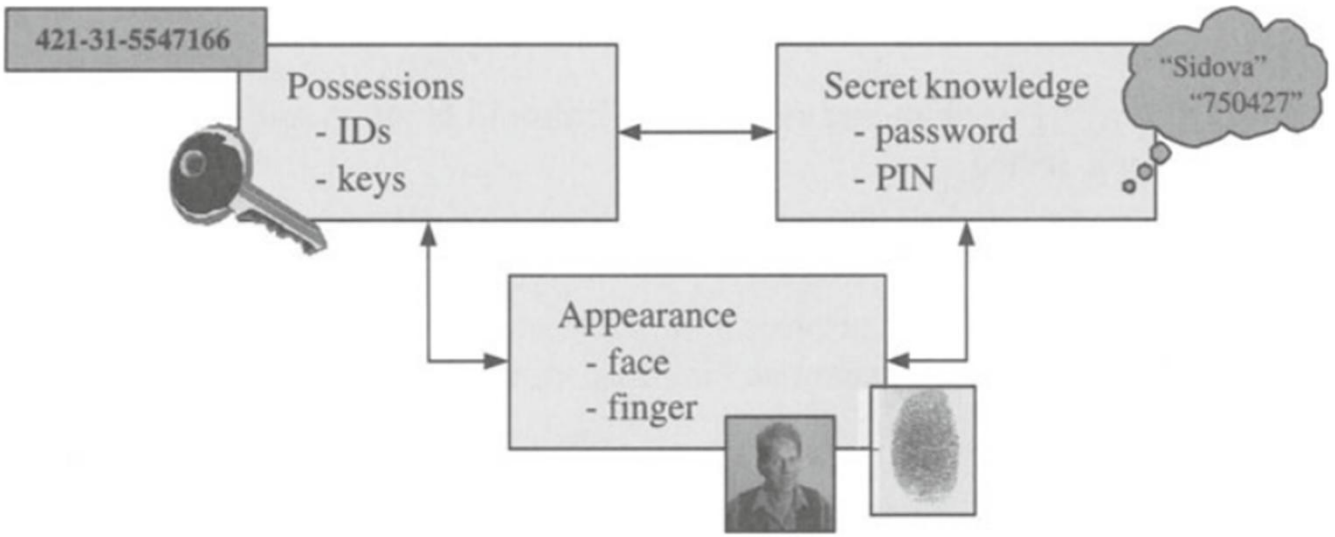
2. *Knowledge:* Pieces of information that are supposed to be kept secret, only known to the rightful person, i.e., passwords and pass phrases.

   Knowledge can include more-or-less personal and private information that might not be secret, such as a user ID, *mother's maiden name*, and *favorite color*.

3. *Biometrics:* Physiological and behavioral appearances or characteristics of individuals that distinguish one person from the next. These are characteristics of the human body and human actions that differentiate people from each other.

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Modes of Authentications



Operation of a biometric system

- data, compares this feature set against the feature set(s) stored in the database.
- A biometric system is essentially a pattern recognition system that acquires biometric data from an individual, extracts a salient feature set from the n action based on the result of the comparison.

Main Modules :

- Sensor module
- Quality assessment and feature extraction module
- Matching and decision-making module
- System database module

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Sensor module

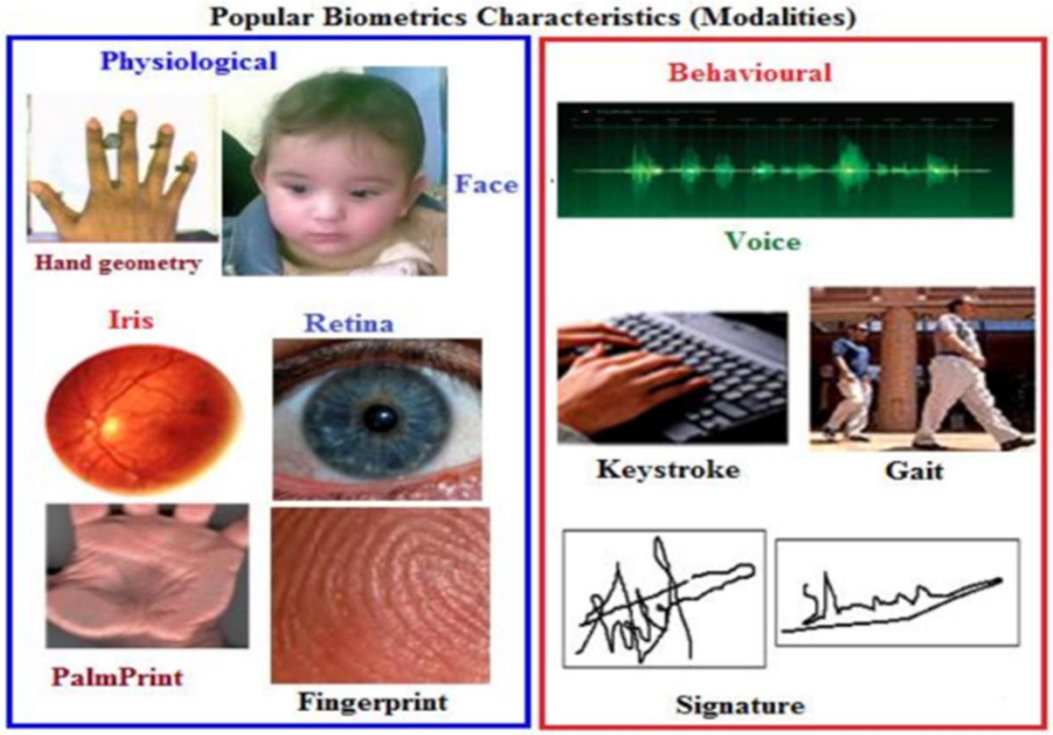
- A suitable biometric reader or scanner is required to acquire the raw biometric data of an individual.

Ex : To obtain fingerprint images, for example, an optical fingerprint sensor may be used to image the friction ridge structure of the fingertip.

- The sensor module defines the human machine interface and is, therefore, pivotal to the performance of the biometric system.
- A poorly designed interface can result in a high failure- to-acquire rate and, consequently, low user acceptability.

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Possible Biometric Traits



Quality assessment and feature extraction module:

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- The quality of the biometric data acquired by the sensor is first assessed in order to determine its suitability for further processing.
- Typically, the acquired data is subjected to a signal enhancement algorithm in order to improve its quality. However, in some cases, the quality of the data may be so poor that the user is asked to present the biometric data again.
- The biometric data is then processed and a set of salient discriminatory features extracted to represent the underlying trait

Quality assessment and feature extraction module:

- For example, the position and orientation of minutia points (local ridge and valley anomalies) in a fingerprint image are extracted by the feature extraction module in a fingerprint-based bio-metric system.

- During enrollment, this feature set is stored in the database and is commonly referred to as a template.

Matching and decision-making module:

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- The extracted features are compared against the stored templates to generate match scores.
- During enrollment, this feature set is stored in the database and is commonly referred to as a template.
- The match score may be moderated by the quality of the presented biometric data. The matcher module also encapsulates a decision making module, in which the match scores are used to either validate a claimed identity or provide a ranking of the enrolled identities in order to identify an individual.

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Biometric characteristics

- Universality: Every individual accessing the application should possess the trait.
- Uniqueness : The given trait should be sufficiently different across individuals comprising the population.
- Permanence: The biometric trait of an individual should be sufficiently invariant over a period of time with respect to the matching algorithm. A trait that changes significantly over time is not a useful biometric.



Biometric characteristics

- Measurability: It should be possible to acquire and digitize the biometric trait using suitable devices that do not cause undue inconvenience to the individual.
  - Furthermore, the acquired raw data should be amenable to processing in order to extract representative feature sets.

  Performance: The recognition accuracy and the resources required to achieve that accuracy should meet the constraints imposed by the application.

  Acceptability: Individuals in the target population that will utilize the application should be willing to present their biometric trait to the system.

Biometric characteristics

Circumvention: This refers to the ease with which the trait of an individual can be imitated using artifacts (e.g., fake fingers), in the case of physical traits, and mimicry, in the case of behavioral traits.

Modes of Biometric System

- Depending on the application, biometric systems can be used in one of two modes:

  - ❑**Verification**

  - ❑ **Identification**

  Verification : (also called authentication) is used to verify a person's identity—that is, to authenticate that individuals are who they say they are

  Identification is used to establish a person's identity— that is, to determine who a person is

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How Biometric Technologies Work

- Biometric identification systems are essentially pattern recognition systems
- They use acquisition devices such as cameras and scanning devices to capture images, recordings, or measurements of an individual's characteristics.
- Computer hardware and software: to extract, encode, store,

and compare these characteristics

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Stages in Biometrics system

- All biometric systems involve similar processes that can be divided into two distinct stages:
- **Enrollment**
- **Verification or Identification**

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Enrollment

- A biometric system is trained to identify a specific person.
- The person first provides an identifier, such as an identity card.
- The biometric is linked to the identity specified on the identification document.
- The distinctive features are located and one or more samples are extracted, encoded, and stored as a reference template for future comparisons
- Template size varies depending on the vendor and the technology.
- Templates can be stored; remotely in a central database or within a biometric reader device itself

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Enrollment

- The quality of the template or templates is critical in the overall success of the biometric application.

- Because biometric features can change over time, people may have to re-enroll to update their reference template.

Enrollment

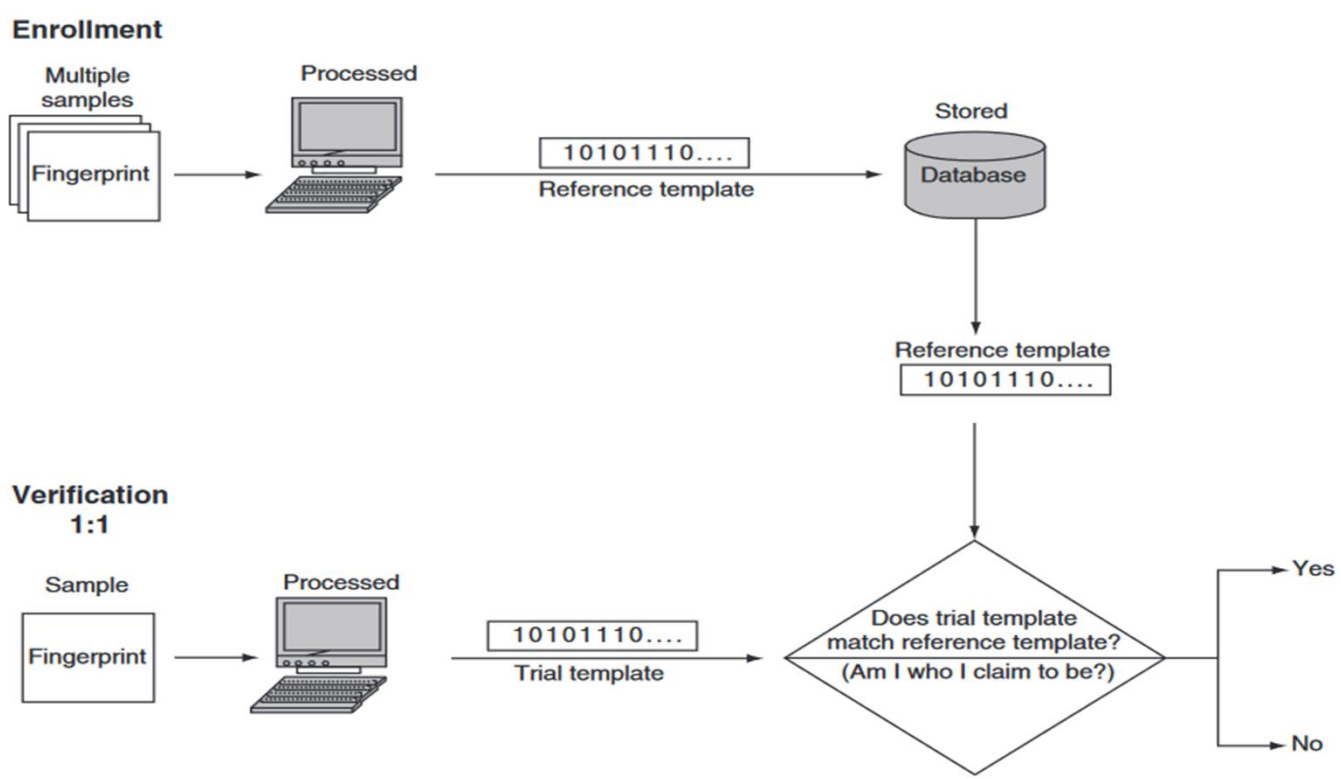


Figure 2-1 *The biometric verification process.*

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Verification Stage in biometric system

- In the verification mode, the system validates a person's identity by comparing the captured biometric data with her own biometric template(s) stored in the system database.

- Verification is often referred to as 1:1 (one-to-one) matching.

- Verification is typically used for positive recognition, where the aim is to prevent multiple people from using the same identity

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Identification Stage in biometric system

- In the identification mode, the system recognizes an individual by searching the templates of all the users in the database for a match.
- Therefore, the system conducts a one-to-many comparison to establish an individual's identity.

- Identification is a critical component in negative recognition applications where the system establishes whether the person is who she (implicitly or explicitly) denies to be. .

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Identification Stage in biometric system

- The purpose of negative recognition is to prevent a single person from using multiple identities.
- Identification may also be used in positive recognition for convenienc

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When Are Verification and Identification Appropriate?

- PC and network security generally employ verification systems; access to buildings and rooms can be effective with either identification or verification systems, though verification is predominant

- Large-scale public benefits programs generally utilize identification systems.

Verification Vs Identification

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- Verification systems are generally faster and more accurate than identification systems.
- Instead of performing hundreds of comparisons against enrolled users, they need only match a person's data against his or her existing data.
- This requires less computing power and decreases the likelihood that the system will match an unauthorized user.
- Nearly all verification systems can ren der a match/no-match decision within less than one second.
- Verification cannot determine whether a given person is present in a database more than once.

Verification Vs Identification

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- Identification systems require more computational power than verification systems, because more comparisons take place before a match occurs some cases, millions of matches

- Identification systems are deployed when verification simply does not make sense (to eliminate duplicate enrollments, for example).
- Although the idea of performing identification in desktop environments may be appealing, the associated reduction in speed and accuracy generally outweigh these modest benefits of eliminating a username or ID.

How Biometric Matching Works

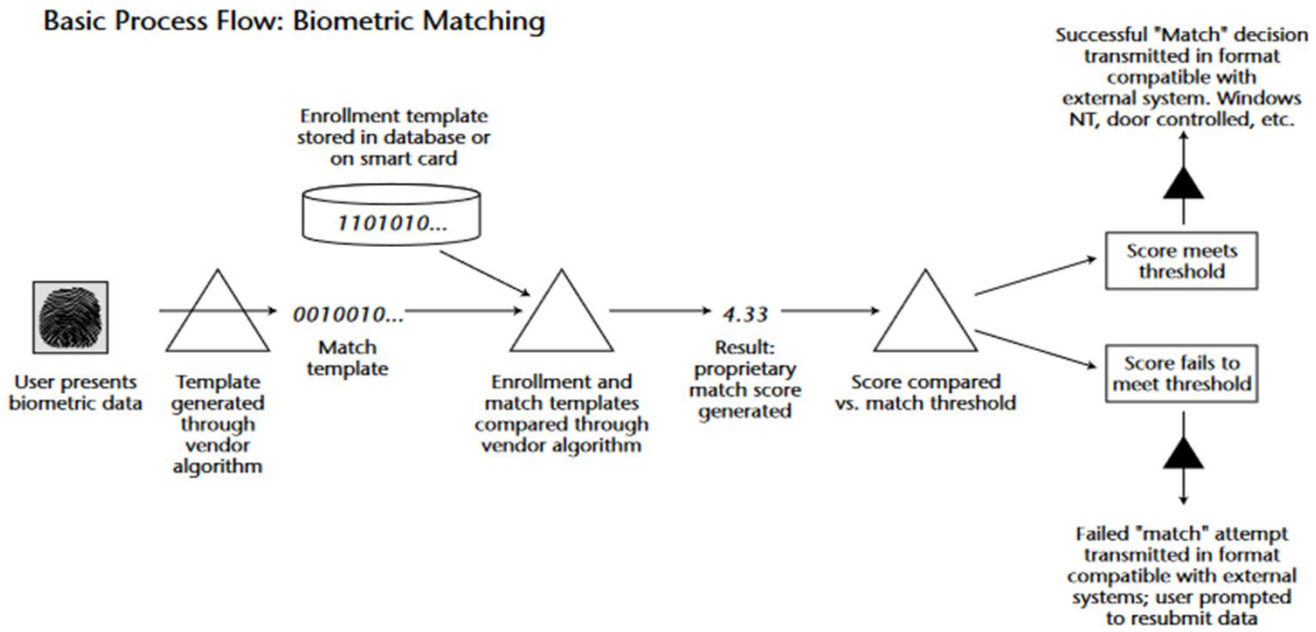


Figure 2.2 Biometric matching: process flow.

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Steps in Biometric matching

- A user initially enrolls in biometric systems by providing biometric data, which is converted into a template.
- Templates are stored in biometric systems for the purpose of subsequent comparison.
- In order to be verified or identified after enrollment, the user provides biometric data, which is converted into a template
- The verification template is compared with one or more enrollment templates

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Steps in Biometric matching

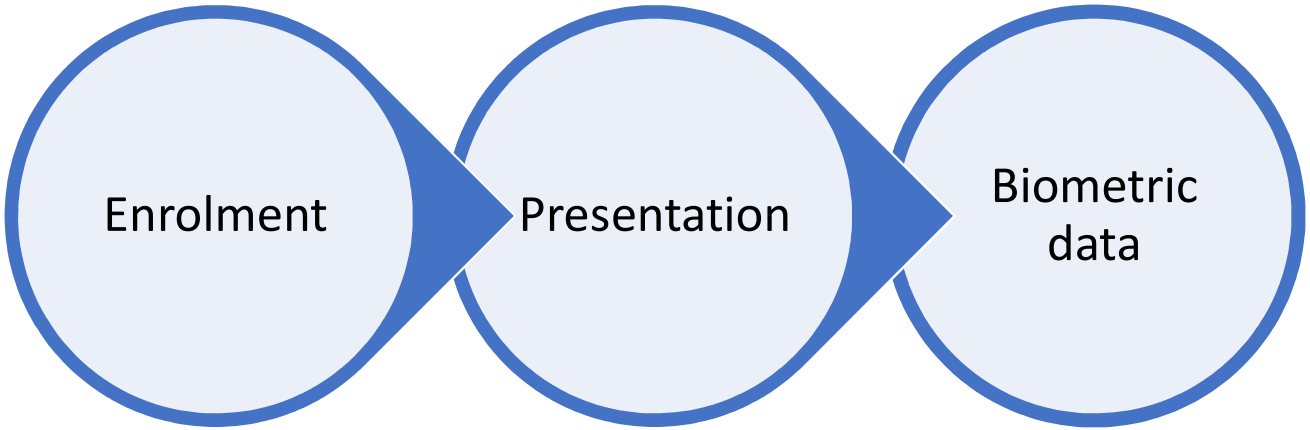
- The result of a comparison between biometric templates is rendered as a score or confidence level, which is compared to a threshold used for a specific technology, system, user, or transaction.

- If the score exceeds the threshold, the comparison is a match, and that result is transmitted
- If the score does not meet the threshold, the comparison is not a match, and that result is transmitted.

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Enrollment and Template Creation

- The following are key terms and processes involved in enrollment and template creation:



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Enrollment

Enrollment : The process by which a user's biometric data is initially acquired, assessed, processed, and stored in the form of a template for ongoing use in a biometric system is called enrollment.

- Subsequent verification and identification attempts are conducted against the template(s) generated during enrollment.
- Enrollment takes place in both 1:1 and 1:N systems
- Quality enrollment is a critical factor in the long-term accuracy of biometric systems
- Low-quality enrollments may lead to high error rates including false match rate and false nonmatch rate.

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Presentation

- **Presentation : It is the process by which a user provides biometric data to an acquisition device—the hardware used to collect biometric data.**
- Depending on the biometric system presentation may require looking in the direction of a camera, placing a finger on a platen, or reciting a passphrase.
- A user may also have to remove eyeglasses or remain still for a number of seconds in order to provide biometric data.
- A user may also have to remove eyeglasses or remain still for a number of seconds in order to provide biometric data

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Biometric Data

- The biometric data users provide is an unprocessed image or recording of a characteristic. This unprocessed data is also referred to as raw biometric data or as a biometric sample.
- Raw biometric data cannot be used to perform biometric matches.
- Biometric systems do not store biometric data—systems use data for template creation.

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Biometric Data

- Depending on the biometric system, a user may need to present biometric data several times in order to enroll.

For example,

- finger-scan systems require the user to place each finger two to four times to gather sufficient data for template creation.

- The enrollment process may also gather data from more than one finger (or iris, or retina) to create multiple enrollment template

Once biometric data has been acquired, biometric templates can be created by a process of feature extraction.

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Data Types and Associated Biometric Technologies

DATA TYPE	BIOMETRIC TECHNOLOGY
Finger-scan	Fingerprint image
Voice-scan	Voice recording
Face-scan	Facial image
Iris-scan	Iris image
Retina-scan	Retina image
Hand-scan	3-D image of top and sides of hand
Signature-scan	Image of signature and record of related dynamics measurements
Keystroke-scan	Recording of characters typed and record of related dynamics measurements

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Feature Extraction

- Defn: The automated process of locating and encoding distinctive characteristics from biometric data in order to generate a template is called feature extraction.
- Feature extraction takes place during enrollment and verification—any time a template is created.
- The feature extraction process includes filtering and optimization of images and data in order to accurately locate features.
- Since quality of feature extraction directly affects a system's ability to generate templates, it is extremely important to the performance of a biometric system.

Templates

- A template is a small file derived from the distinctive features of a user's biometric data, used to perform biometric matches.
- Biometric systems store and compare biometric templates, not biometric data

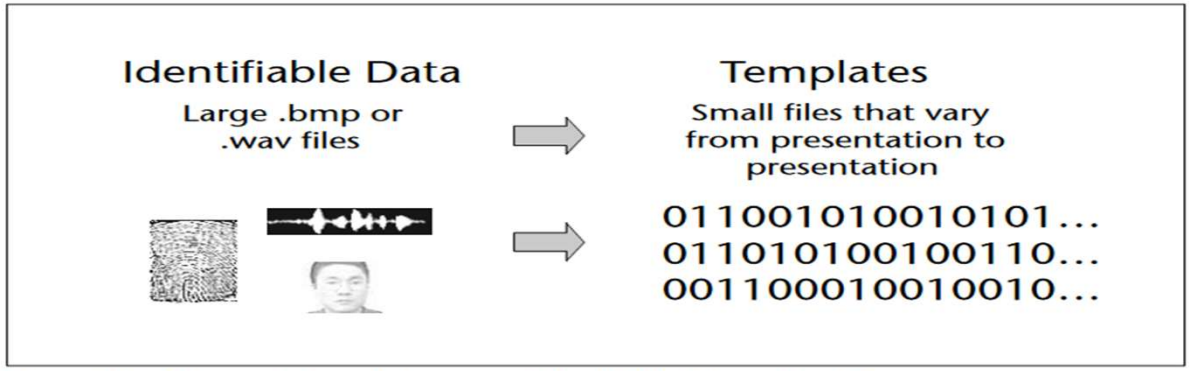
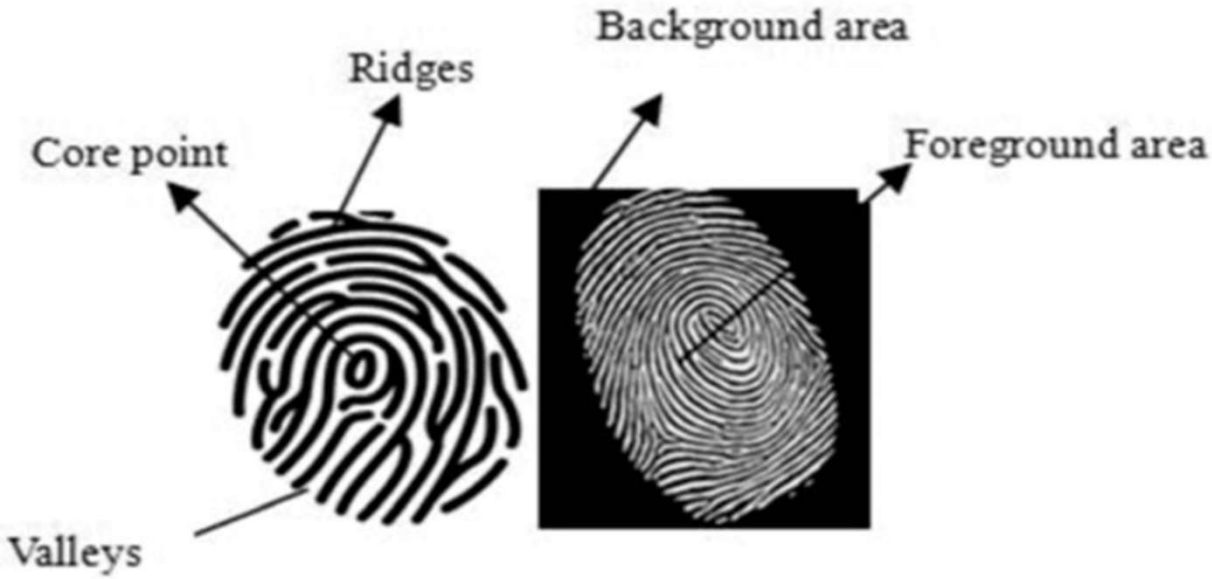


Figure 2.3 Biometric templates versus identifiable biometric data.


Background area
Ridges
Core point
Foreground area
Valleys


The Authentication Process
STOP
Screening
Scanning
Feature extraction
Association
Decision

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Types of Templates

Enrolment Templates

- Enrollment templates are created upon the user's initial interaction with a biometric system and are stored for usage in future biometric comparisons

Match templates

- Match templates are generated during subsequent verification attempts or identification attempts, compared to the stored template, and generally discarded immediately after the comparison

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Biometric Matching

- **The comparison of biometric templates to determine their degree of similarity or correlation is called matching.**
- The process of matching biometric templates results in a score, which, in most systems, is compared against a threshold.
- If the score exceeds the threshold, the result is a match; if the score falls below the threshold, the result is a nonmatch.

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Biometric Matching

- The matching process involves the comparison of a verification template, created when the user provides biometric data, with the enrollment template(s) stored in a biometric system.

- In verification systems, a verification template is matched against a user's enrollment template or templates (a user may have more than one biometric template enrolled

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Scoring

- **Defn:** Biometric match/no-match decisions are based on a score—a number indicating the degree of similarity or correlation resulting from the comparison of enrollment and verification templates.

- There is no standard scale used for biometric scoring:
- Some biometric systems employ a scale of 1 to 100; others use a scale of -1 to 1
- These scores can be carried out to several decimal points and can be logarithmic or linear. Scoring systems vary not only from technology to technology, but from vendor to vendor.

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Threshold

- **Definition** : Once a score is generated, it is compared to the verification attempt's threshold.
- A threshold is a predefined number, generally chosen by a system administrator, which establishes the degree of correlation necessary for a comparison to be deemed a match.
- Thresholds can vary from user to user, from transaction to transaction, and from verification attempt to verification attempt.
- Systems can be either highly secure or not secure at all, depending on their threshold settings.

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Decision

- The result of the comparison between the score and the threshold is a decision.
- The decisions a biometric system can make include match, nonmatch, and inconclusive, although varying degrees of strong matches and nonmatches are possible.

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Accuracy in Biometric Systems

- The key performance metrics in biometrics are
  - False match rate, (FMR)
  - False non- match rate (FNMR)
  - Failure-to-enroll rate (FER)

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False match rate, (FMR)

- Definition : A biometric solution's false match rate is the probability that a user's template will be incorrectly judged to be a match for a different user's template.
- False matches may occur because two people have similar enough biometric characteristics
- The false match rate is often referred to as the false acceptance rate (FAR).

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False match rate, (FMR)

- False match rates can be reduced by adjusting thresholds that adjust the level of correlation necessary for two templates to be judged a match.
- False match rate and false nonmatch rate are inversely related and must always be assessed.
- Ex: A system with a false match rate of 0 percent, but a false nonmatch rate of 50 percent, is secure but unusable.

False match rate, (FMR)

- The false match rate is often referred to as the false acceptance rate (FAR).
- The term false acceptance rate assumes that the result of a successful match is that the user is accepted into a building, an application, or a resource, which is often the case.
- The false match rate is often referred to as the false acceptance rate (FAR).

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When Are False Matches Acceptable?

- A false match does not always mean that the biometric system has failed or needs threshold adjustment.
- Real Scenario :Think of a video security system used by a casino to identify known card counters.

Solution :

- Without Biometrics : security personnel need to memorize the faces of every known card counter, then look at every face at every blackjack table.

When Are False Matches Acceptable?

Solution :

- With Biometrics : facial-scan software can scan the images provided by security cameras, locate faces, and rapidly perform 1:N identify cation against databases of card counters.
- The facial-scan system will be programmed to notify or flag any faces that come at all close to any images in the card counter database, at which point the security personnel can determine whether the two faces—one in the database, one from the live security camera—are a match. Even if 9 out of 10 matches are incorrect, the facial-scan system is performing a valuable function by alerting staff to possible matches and reducing overall manual effort.

False Match Rate in Large-Scale Identification Systems

- FMR are often different for large-scale identification systems. Because over a billion matches might take place on a given day

- large-scale identification requires much lower false match rates than would commonly be the case in verification systems.
- In particular, negative identification systems—those designed to ensure that an individual's biometric data is not present more than once in a database require low false match rate

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False Nonmatch Rate(FNMR)

- (FNMR) is the probability that a user's template will be incorrectly judged to not match his or her enrollment template.
- In most cases, a false nonmatch means that an authorized user is locked out of a system, incorrectly denied access to a facility or resource.
- A solution's false nonmatch rate is often referred to as false rejection rate (FRR).

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False Nonmatch Rate(FNMR)

- False nonmatches occur because there is not a sufficiently strong correlation between a user's verification and enrollment templates.

Main Reason for FNMR:

- **Changes in a user's biometric data**
- **Changes in how a user presents biometric data**
- **Changes in the environment in which data is presented**

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Changes in a User's Biometric Data

- Many biometric technologies are based on characteristics that can change between enrollment and verification, especially as a substantial amount of time elapses.
- Examples :
- Voice-scan systems, for example, do not perform as well when users have sore throats;
- facial-scan systems perform poorly if a user has experienced drastic changes in facial hair or weight.

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Changes in a User's Biometric Data

- Even more stable characteristics such as fingerprints can change over time, as scars, aging, and general wear and tear lower the quality of the data.
- Technologies somewhat susceptible to changes in underlying biometric data include facial-scan and hand-scan; eye-based biometrics, iris-scan, and retinal scan are less susceptible. Long-term robustness of finger-scan solutions varies too widely from vendor to vendor to make a technology-wide assessment.

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Changes in User Presentation

- Changes in User Presentation Even if the biometric data has not changed in any way, the manner in which a user presents data to a biometric system can lead to substantial false rejection rates.
- These changes are not necessarily attributable to user carelessness.
- Finger-scan systems may advise users to move a finger up or down;
- voice-scan systems may advise users to speak more softly or loudly.
- Behavioral biometrics, in particular, are subject to changes in user presentation.

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Changes in Environment

- changes in authentication environment.
  - Variations in **background lighting and composition, noise, and even temperature can lead to increased false non- match rates.**

Example :

Enrolling on a landline telephone, for example, and attempting to verify via mobile phone is likely to lead to high FNMR.

Factors Affecting False Nonmatch Rates

TECHNOLOGY	SUSCEPTIBILITY TO CHANGES IN BIOMETRIC DATA	SUSCEPTIBILITY TO CHANGES IN USER PRESENTATION	SUSCEPTIBILITY TO ENVIRONMENTAL CHANGES
Finger-scan	*Low to moderate.* Some user groups are prone to changes in fingerprint quality; for most users, the fingerprint remains stable and usable.	*Moderate.* Variation in angle, pressure, and position can lead to high FNMR.	*Low.* Some solutions are susceptible to light and temperature changes, but this is rare.
Facial-scan	*Moderate to high.* Changes in facial hair and hairstyle, as well as drastic changes in weight, can increase FNMR.	*Low to moderate.* In a controlled environment, user presentation should not vary. Otherwise, user is responsible for quality of presentation. Removal of glasses and hats can also increase FNMR.	*Moderate to high.* Changes in intensity and angle of lighting, as well as background composition, can be highly problematic for facial-scan.
Voice-scan	*Low to moderate.* Illnesses can affect the voice to the point where FNMR increases.	*Moderate to high.* Volume of speech, inflection, and duration must be consistent to ensure verification.	*Moderate to high.* Background noise and quality of acquisition device (microphone, telephone) can increase FNMR.

Factors Affecting False Nonmatch Rates

Iris-scan	*Low.* The iris is highly stable and, short of eye trauma, iris patterns do not change.	*Moderate to high.* Users must be positioned correctly in order for high-quality image to be captured.	*Low.* Lighting does not affect the technology, as it uses infrared illumination.
Signature-scan	*Moderate to high.* Data and presentation cannot be effectively separated. For some users, the signature is highly susceptible to variation, especially over time. Signing in different position (sitting vs. standing) may impact performance.		*Low.* Technology largely unaffected by environment.
Hand-scan	*Moderate.* Injuries, swelling, water retention can increase likelihood of false nonmatching.	*Low to moderate.* Design of technology facilitates correct placement, but placement itself can be challenging for some users.	*Low.* Technology largely unaffected by environment.

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TECHNOLOGY	SUSCEPTIBILITY TO CHANGES IN BIOMETRIC DATA	SUSCEPTIBILITY TO CHANGES IN USER PRESENTATION	SUSCEPTIBILITY TO ENVIRONMENTAL CHANGES
Retina-scan	*Low.* Aside from dislocation due to age-related eye diseases, retinal patterns are stable.	*High.* User presentation is extremely challenging, leading to atypically high FNMR.	*Low.* Technology largely unaffected by environment.
Keystroke-scan	*Moderate to high.* Data and presentation cannot be effectively separated. Typing patterns easily change, especially for poor typists. Effect of different keyboard types (click vs. nonclick, ergonomic keyboards) unknown.		*Low.* Technology largely unaffected by environment.

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Failure-to-Enroll (FTE) Rate

- A system's failure-to-enroll (FTE) rate represents the probability that a given user will be unable to enroll in a biometric system.
- FTEs occur when users have insufficiently distinctive or replicable biometric data or when the design of the biometric solution is such that providing consistent data is difficult.
- High failure-to-enroll rates can be particularly problematic for a biometric system, as users unable to enroll must verify through another biometric technology or authentication method.

Failure-to-Enroll by Technology

TECHNOLOGY	CAUSES OF FTE
Finger-scan	Low-quality fingerprints (scarred, faint, no breaks in ridges)
	Inconsistent submission (angle and pressure of placement)
	Difficult submission process (sliding, swiping finger)
Facial-scan	Glasses, hats, headwear
	Poor lighting
	Background composition
Voice-scan	Background noise
	Quality of acquisition device
	Inconsistent vocal samples (volume, duration, inflection)
Iris-scan	Difficult submission process (lining up eye with guides, gauging correct distance)
	Glasses (cause reflection)
Signature-scan	Inconsistent signatures
Hand-scan	Inability to position hand according to guide pegs
Retina-scan	Difficult submission process (must remain still for several seconds for each acquisition, multiple acquisitions required)
Keystroke-scan	Inconsistent typing patterns

Relation of Failure-to-Enroll to False Nonmatch Rate

- False match rates and false nonmatch rates are inversely related: Decrease one rate and the other rate increases, and vice versa.

  FMR= 1/FNMR

- Adjusting system settings to reduce FTE rates can also affect false nonmatch rates.
- By reducing the number of distinctive or replicable features required for enrollment
- FTE rates can be lowered—marginal images and data can then be enrolled.

Derived Metrics

- Equal Error Rate (EER): The equal error rate (EER), also referred to as the crossover rate, is the rate at which the FNMR is equal to the FMR.
- it represents the accuracy level at which likelihood of a false match is the same as the likelihood of a false non- match.
- Equal error rate is commonly used as a representation of overall system accuracy, as it is a general indicator of a system's resistance to break-ins and ability to match templates from authorized users

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Derived Metrics

- EER is generated from single false match and single false nonmatch rates, neither of which is reflective of system operation in the real world.
- EER does not incorporate failure-to-enroll rates.
- A system might have an extremely low EER but an FTE rate of 15 percent.

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Ability-to-Verify (ATV) Rate

- A more valuable derived metric is the ability-to-verify (ATV) rate.
- ATV is a combination of the failure-to-enroll and false nonmatch rates, and indicates the overall percentage of users who will be capable of authenticating on a daily basis.

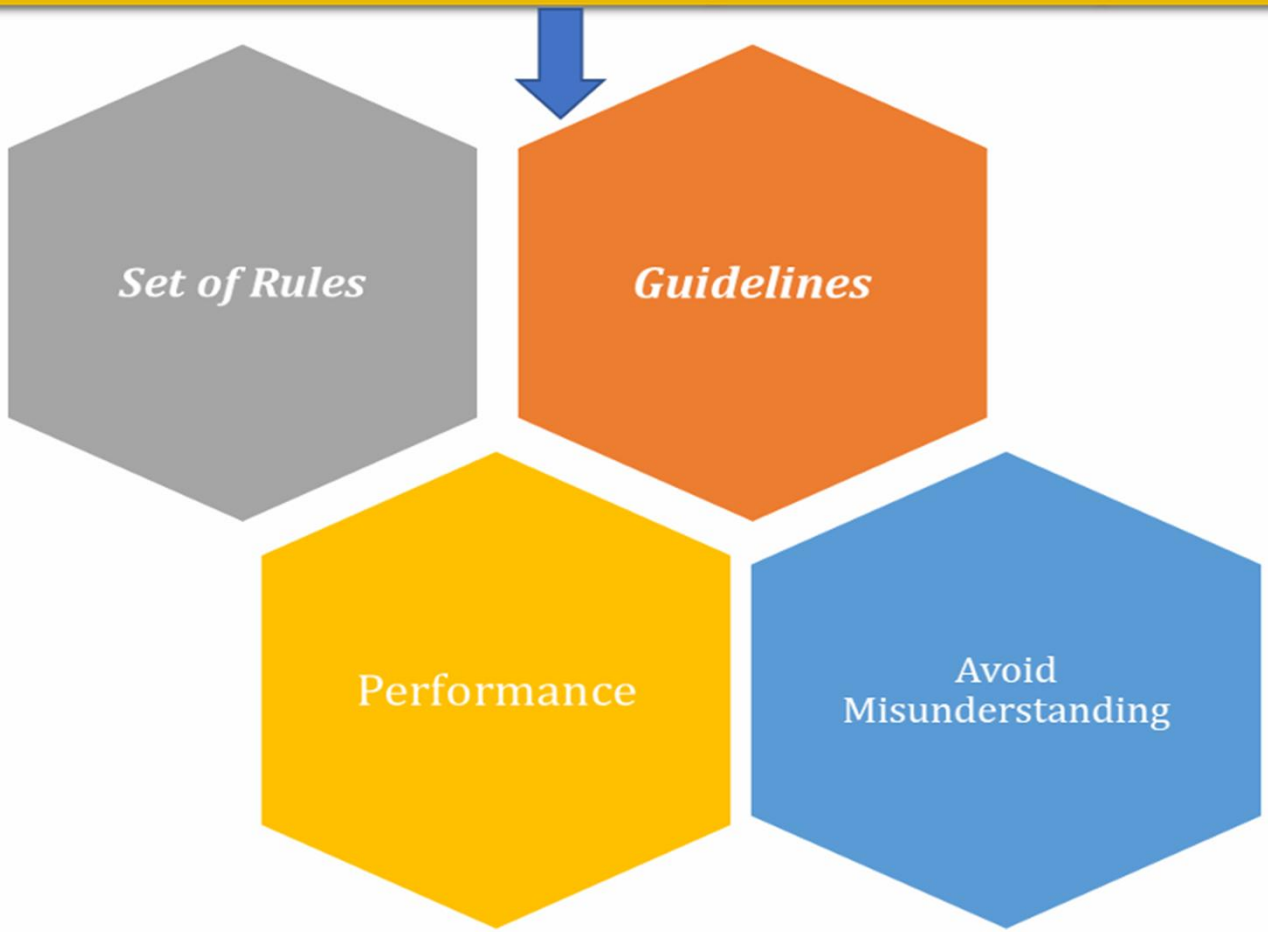
$$ATV = (1 - FTE)(1- FNMR)$$

Designing privacy sympathetic biometric systems

Different biometric standards,

- The Biometric Consortium website defines standards as “a general set of rules to which all complying procedures, products or research must adhere.”
- Standards play a role in everyday life by establishing the size, configuration, or protocol of a product, process, or system.

Standards: "*a document, established by consensus that provides:*



- Standards for the collection, storage, and sharing of biometric data are of utmost importance to government and private systems.
- International Organization for Standardization (ISO)/International Electrotechnical Commission (IEC) Guide 2:2004 defines a standard as “a document, established by consensus that provides rules, guidelines or characteristics for activities or their results.”
- The Biometric Consortium website defines standards as “a general set of rules to which all complying procedures, products or research must adhere.”
- Standards play a role in everyday life by establishing the size, configuration, or protocol of a product, process, or system.
- Standards specify performance of products or personnel and also define terms so that there is no misunderstanding among those using the standards

Biometric standards specify:

- formats for the interchange of biometric data
- common file formats that provide platform independence
- application program interfaces and application profiles
- performance metric definitions and calculations
- approaches to test performance

Technical Interfaces:

- specify interfaces and interactions between biometric components and subsystems, including the possible use of security mechanisms to protect stored data and data transferred between systems. Examples include ANSI INCITS 358-2002 BioAPI Specification v1.1 and ANSI INCITS 398-2005 [NISTIR 6529-A] Common Biometric Exchange File Format (CBEFF).

Data Interchange Formats:

- Examples include ANSI INCITS 377-2004 Finger Pattern Based Interchange Format, ANSI INCITS 378-2004 Finger Minutiae Format for Data Interchange, and ANSI INCITS 379-2004 Iris Image Interchange Format.

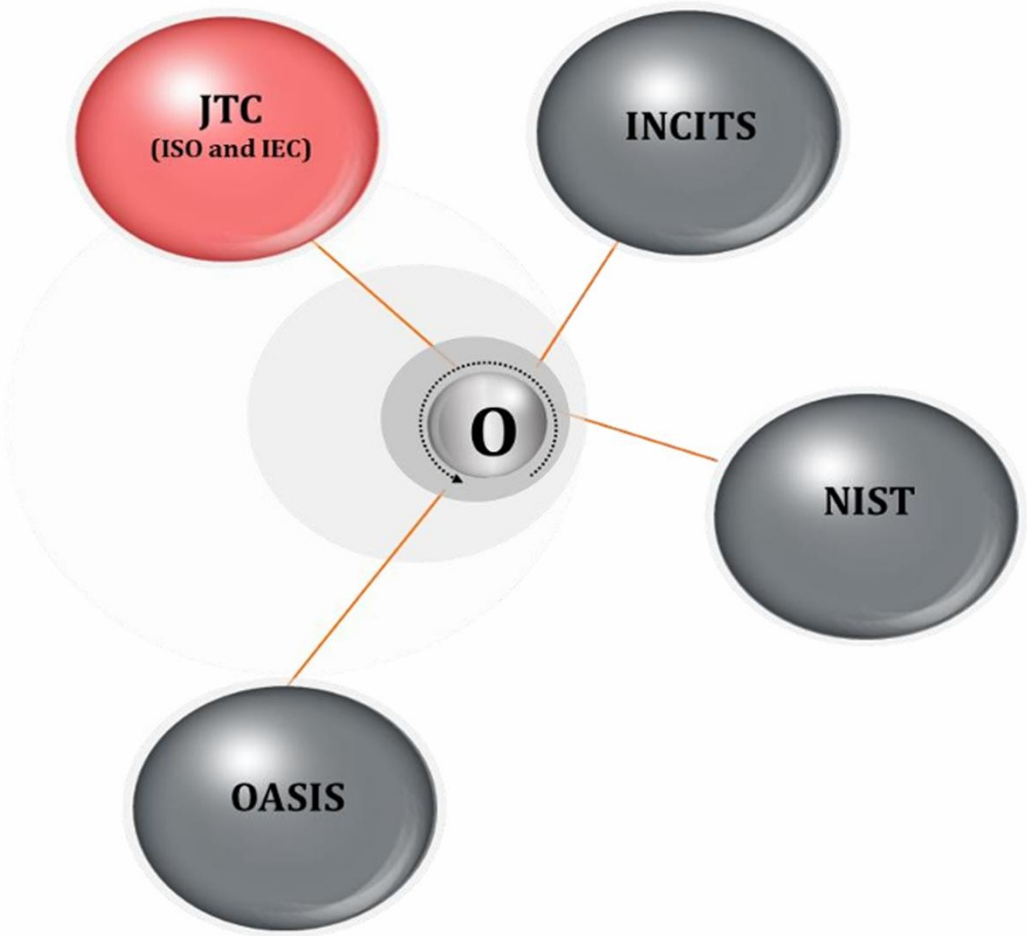
Application Profile Standards:

- Examples include ANSI INCITS 383-2003 Biometrics-Based Verification and Identification of Transportation Workers, and ANSI INCITS 394-2004 Data Interchange and Data Integrity of Biometric-Based Personal Identification for Border. Management.

Performance Testing and Reporting :

- Examples include ANSI INCITS 409.1-2005 Biometric Performance Testing and Reporting Part 1 - Principles Framework; ANSI INCITS 409.2-2005 Biometric Performance Testing a Reporting Part 2 - Technology Testing Methodology; and ANSI INCITS 409.3-2005 Biometric Performance Testing and Reporting Part 3 - Scenario Testing Methodologies.

Major Organizations involved in drafting Standards



- **INCITS**: InterNational Committee for Information Technology Standards (INCITS)
- **NIST**: National Institute of Standards and Technology
- **JTC**: Joint Technical Committee 1 (JTC 1)/Subcommittee 37 (SC 37). ISO and IEC Joint Technical Committee
- **OASIS**: Organization for the Advancement o Structured Information Standards.

Working groups of JTC

Working Group	Responsibilities
WG1 Harmonized Biometric Vocabulary	develops standardized definitions for biometric vocabulary terms
WG2 Biometric Technical Interfaces	develops international standards for BioAPI and CBEFF, as well as a number of other related projects.
WG3 Biometric Data Interchange Formats	develops international versions of the biometric data interchange format standards.
WG4 Biometric Functional Architecture and Related Profiles	develops international biometric profile standards to support biometric interoperability for applications.
WG5 Biometric Testing and Reporting	develops international standards for biometric performance testing and reporting.
WG6 Cross jurisdictional and Societal Aspects	currently developing an international technical report on privacy concerns and other social concerns related to biometric standards.